The Cosmological Constant

Why It Isn't Zero But Nearly So

UGP Physics Tutorial Series

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Claim-strength taxonomy (used throughout)

CatAL Machine-certified in Lean 4, zero sorry, zero custom axioms.

CatAD Full analytic derivation, computationally confirmed.

CatA Computationally verified; analytic or Lean cert pending.

CatB Plausible / partially supported.

Contents

UGP Physics Tutorial Series

Prerequisites (read first): *Gravity from Description-Length Minimization* · *The Born Rule as a Theorem*

Next tutorials: *$\mathbb{Z}_7$ Defect Cosmology*

1 What Is Dark Energy?

The one-line answer

Dark energy is whatever is causing the expansion of the universe to *speed up* instead of slow down. It is modelled as a constant background energy density built into the fabric of space itself.

1.1 The universe is not just expanding — it is accelerating

In 1998, two independent teams measuring the brightness of distant stellar explosions (Type Ia supernovae) made an unexpected discovery: the universe is expanding *faster* today than it was in the past. This was a shock. Gravity pulls matter together, so the expansion should be slowing down as it does when a ball is thrown upward. Instead it is speeding up.

Something must be providing a repulsive push — a kind of anti-gravity acting on cosmic scales. The simplest way to model this is to add a constant energy density Λ (measured in energy per unit volume) to Einstein's equations for gravity. When this constant is positive, it acts like a negative pressure: space itself has energy, and that energy pushes space apart faster and faster.

We call this energy **dark energy**, and we call the constant Λ the **cosmological constant**.

1.2 How much dark energy is there?

Cosmologists parameterize the energy budget of the universe as fractions of the critical energy density ρ_{crit} — the density at which the universe is geometrically flat. The dark energy fraction is written Ω_Λ.

Planck satellite measurements of the cosmic microwave background (CMB, Planck 2018) give:

$$\Omega_\Lambda = 0.6889 \pm 0.0056.$$

Roughly **69%** of the universe's total energy is dark energy. About 26% is dark matter, and only 5% is ordinary matter (atoms, stars, planets, you).

Energy budget of the universe

Component	Fraction
Dark energy (Ω_Λ)	$\approx 69\%$
Dark matter (Ω_{DM})	$\approx 26\%$
Ordinary (baryonic) matter	$\approx 5\%$

2 The Cosmological Constant Problem

Key idea

Quantum field theory predicts that empty space should be seething with virtual-particle fluctuations that contribute an enormous vacuum energy. The prediction misses the observed value by a factor of 10^{120} — the worst mismatch between theory and experiment in the history of science.

2.1 Why empty space should be energetic

Quantum mechanics says that the vacuum — empty space with no particles in it — is not actually empty. By the Heisenberg uncertainty principle, even the lowest-energy state of every quantum field has fluctuating virtual particles constantly appearing and disappearing. These fluctuations contribute energy to the vacuum.

When you add up the contributions from all quantum fields up to the energy scale where our best theories are known to work (the Planck scale, $M_{\text{Pl}} \approx 10^{18}$ GeV), you get a predicted vacuum energy density of roughly:

$$\rho_\Lambda^{\text{QFT}} \sim M_{\text{Pl}}^4 \approx 10^{76}\ \text{GeV}^4.$$

2.2 Why this is the wrong answer by an extraordinary margin

The observed dark energy density, back-calculated from Ω_Λ, is:

$$\rho_\Lambda^{\text{obs}} \approx 10^{-47}\ \text{GeV}^4.$$

The ratio is:

$$\frac{\rho_\Lambda^{\text{QFT}}}{\rho_\Lambda^{\text{obs}}} \approx 10^{120}.$$

In plain English: the standard theory overshoots by a factor of 10^{120}.

The worst fine-tuning problem in physics

To agree with observation, the bare cosmological constant and the quantum vacuum contributions must cancel to 120 decimal places, with a tiny nonzero remainder. Nobody knows why this enormous cancellation occurs, or why the remainder is the specific nonzero value we observe. This is called the **cosmological constant problem**, often described as the largest discrepancy between theory and observation in all of physics.
There are two separate puzzles:

1. **The small-but-nonzero problem:** Why is Λ so incredibly small but not zero?
2. **The fine-tuning problem:** Why do enormous contributions cancel so precisely — to 120 decimal places?

2.3 Why $\Lambda = 0$ alone would not be satisfying

One might hope that some symmetry simply forces $\Lambda = 0$ exactly, making the problem go away. But the universe *tells* us through observation that $\Lambda \neq 0$. So the symmetry cannot be perfect, or something breaks it at just the right level. Both sub-problems must be addressed simultaneously.

3 Level 1: Classical $\Lambda = 0$ Exactly

Key idea

In the GTE framework the physical substrate has exactly **seven** ground states. These seven vacua are identical by symmetry — they are in perfect equilibrium. Because they are all equally weighted, their contributions to the vacuum energy cancel exactly. The classical cosmological constant is zero by symmetry.

3.1 The physical substrate and its seven vacua

The GTE framework derives the Standard Model from a single foundational structure: the field Φ_{MDL}, a quantum field with a mod-7 ($\mathbb{Z}_7$) symmetry built into its potential V. Just as a ball rolling on a landscape can sit in any local valley, Φ_{MDL} can sit in any of its vacuum states. The $\mathbb{Z}_7$ symmetry gives the potential exactly **seven** identical minimum-energy valleys:

$$V(\phi_0) = V(\phi_1) = V(\phi_2) = \cdots = V(\phi_6) = 0.$$

Each ground state has zero potential energy. All seven are related by a discrete shift symmetry: rotating the field by one step ($\phi \to \phi + 2\pi/7$) maps each vacuum to the next.

3.2 Symmetry kills the classical vacuum energy

Because all seven vacua are geometrically identical and related by an exact symmetry, they are equally probable. When a physical quantity is computed by averaging over all vacua (as one must in a theory with degenerate ground states), the contributions from the seven sectors *cancel exactly*. The classical vacuum energy is zero — not by fine-tuning, but by symmetry.

This is analogous to three identical weights placed symmetrically on a scale: the torques cancel exactly, not because someone adjusted them, but because of the symmetry of their placement. No adjustment of any parameter is required.

The Lean-certified chain (CatAL)

Three zero-sorry Lean 4 theorems certify this result:

- `z7_vacuum_sectors_equiprobable` — the seven $\mathbb{Z}_7$ vacua are equiprobable under any $\mathbb{Z}_7$-symmetric observable.
- `phimdl_tmunu_vacuum_zero` — the stress-energy tensor vanishes at every vacuum (zero potential energy).
- `z7_vacuum_energy_mass_independent` — the vacuum energy is a topological invariant; it does not depend on the mass parameter.

Conclusion (CatAL): The classical contribution to Λ is exactly zero. There is no classical vacuum energy to cancel.

3.3 What this resolves — and what it does not

Level 1 eliminates the fine-tuning problem in its classical form: since there is no classical Λ to begin with, there is nothing to cancel. However, quantum fluctuations (virtual-particle loops) can still contribute. Level 2 addresses those.

4 Level 2: Why Quantum Corrections Are Tame

Key idea

The GTE physical substrate encodes three-dimensional space on one-dimensional tapes. This holographic architecture suppresses one-loop vacuum fluctuations by a factor of $\approx 7 \times 10^{-83}$ compared to the standard QFT prediction — a suppression of 83 orders of magnitude.

4.1 The holographic mode count

In standard quantum field theory, a box of volume L^3 contains $\sim L^3$ independent modes (degrees of freedom) for each field. Adding up the vacuum energy of all these modes gives the enormous $\rho_\Lambda \sim M_{\mathrm{Pl}}^4$.

In the GTE framework, three-dimensional space emerges from three one-dimensional tapes (the Chiral Minkowski Cellular Automaton, three tapes encoding 3+1D spacetime via the Dimensional Protocol Principle, CatAL). A region of linear size L has only $3L$ independent modes — not L^3. This *holographic mode count* is the GTE structural fact that tames the quantum vacuum.

4.2 The suppression factor

The ratio of the one-loop quantum correction to the tree-level contribution is (CatAD, machine-certified):

$$\frac{\delta\rho_{\mathrm{loop}}}{\rho_{\mathrm{tree}}} \approx \frac{3H_0^2}{M_{\mathrm{kink}}^2} \approx 7.4 \times 10^{-83},$$

where H_0 is the Hubble constant and M_{kink} is the GTE kink mass. For comparison, the standard QFT estimate gives a ratio of $\approx 10^{+40}$. The GTE holographic architecture suppresses the loop contribution by $83 + 40 = 123$ orders of magnitude relative to the naive QFT value.

4.3 What remains

After Level 1 removes the classical contribution and Level 2 suppresses the quantum loops by 10^{-83}, is the cosmological constant simply zero? No. The universe tells us $\Omega_\Lambda \approx 0.69$. Level 3 explains why a small but nonzero contribution is *unavoidable* — not from a quantum calculation, but from the logic of self-reference.

5 Level 3: Why $\Omega_\Lambda \neq 0$

Key idea

A self-contained universe — one that carries its own complete description — must maintain an irreducible "ledger" of undecidable self-references. This ledger has a physical cost: a nonzero energy density. That energy density *is* the dark energy.

5.1 The PSC requirement: the universe must describe itself

The GTE framework is built on a foundational requirement: a consistent universe must be **perfectly self-contained** (PSC). This means the universe cannot get its laws from the outside; its description must be internal. Formally, the substrate must satisfy *Reflexive Closure (RC)*: the substrate encodes its own complete description.

This is a powerful constraint. It says the universe is, in a precise mathematical sense, a self-describing system.

5.2 Gödel's shadow over the vacuum

Here is where logic enters physics. Gödel's incompleteness theorem (1931) established that any sufficiently powerful formal system that tries to describe itself completely will inevitably encounter statements that it cannot decide are true or false. The self-referential diagonal construction always produces an undecidable fragment.

The same is true for the physical substrate. When the substrate tries to encode its own complete description (as PSC Reflexive Closure requires), it encounters an irreducible undecidable fragment. This is not a failure of the theory; it is a *structural necessity*. It is certified in Lean 4:

Theorem: Physical Incompleteness (CatAL)

PSC Reflexive Closure $\Rightarrow$ the self-referential fragment of the substrate is not Turing-computable.

`asr_rt_not_computable` (zero sorry, from NEMS Paper 11)

5.3 The adjudication floor $D_{\text{res}} > 0$

The undecidable self-referential fragment must be physically *adjudicated* — the universe must somehow process and record its own undecidable self-referential content. This is not optional: Reflexive Closure requires it. The irreducible cost of that adjudication is captured by the **adjudication floor** D_{res}.

The key theorem (certified in Lean 4) establishes that D_{res} is strictly positive:

Theorem: The PSC-Adjudication Theorem (CatAL)

PSC axioms $\Rightarrow$ physical incompleteness $\Rightarrow$ $D_{\text{res}} > 0$ $\Rightarrow$ $\Omega_\Lambda \neq 0$.

Zero sorry, zero custom axioms. The cosmological constant is the necessary consequence of the substrate being a PSC-consistent self-referential system.

`incompleteness_implies_nonzero_omega_lambda` (CatAL)

5.4 A plain-English reading: the universe's self-memory overhead

The adjudication floor has a vivid intuitive reading. Think of the universe as a computer that must continuously maintain a record of its own complete state. Some fraction of that record is undecidable — it cannot be compressed or simplified away. This irreducible fraction represents an unavoidable "computational overhead" that must be paid at every moment.

That overhead is the cosmological constant. Dark energy, on this picture, is not a mysterious substance. It is the irreducible cost of the universe being self-consistent and self-describing. The universe must remember itself, and that act of self-memory costs energy. Formally, D_{res} enters the PMDL action as a positive residual term in the Lagrangian density; via the Friedmann equation $3H_0^2\,\Omega_\Lambda = 8\pi G\,\rho_\Lambda$, this constant energy density maps directly to $\Omega_\Lambda \neq 0$ with no free parameter.

Why the same incompleteness drives quantum measurement

The same diagonal undecidability that forces $D_{\rm res} > 0$ also governs individual quantum measurement outcomes. The Physical Incompleteness Theorem (Ch. 10 of P48 [1]) shows that the universe cannot predict all of its own measurement outcomes algorithmically. The incompleteness that makes quantum randomness irreducible is the *same* incompleteness that prevents the cosmological constant from being zero. The two phenomena share a single root cause.

6 The Bracket: $\Omega_\Lambda \in [3\pi/14,\ 0.6899]$

Key idea

GTE derives Ω_Λ from two independent structural routes. The first gives an upper bound (the information capacity of the universe's self-description); the second gives a lower bound (the minimum physical cost of the hardware that carries that description). Together they bracket the observed Planck 2018 value.

6.1 Route 1: The census ceiling ($\Omega_{\rm PSC} = 0.6899$)

The adjudication floor $D_{\rm res}$ involves the substrate encoding its own gauge-sector description. The gauge sector has an orbit structure with $2000/3 = D^2 N_{\rm fam}^{N_{\rm gen}}/N_{\rm gen} = 4^2 \cdot 5^3/3$ discrete orbit classes. Each class must be encoded in binary at a cost of $\log_2(2000/3)$ bits.

The *maximum possible* contribution (at full census capacity — counting every orbit class) gives the ceiling:

$$\Omega_{\rm PSC} = \frac{\ln 2}{3\pi} \cdot \log_2\left(\frac{2000}{3}\right) = 0.6899. \tag{1}$$

Every numerical factor traces to a GTE structural constant certified independently of Ω_Λ: $D = 4$ (spacetime dimension, CatAL), $N_{\rm fam} = 5$ ($\mathbb{Z}_7$ orbit, CatAL), $N_{\rm gen} = 3$ (PSC, CatAL), $|\mathbb{Z}_7| = 7$ (MDL, CatAL). Lean cert: `psp_epoch_selection_master` (CatAL).

This is the **capacity ceiling**: the realized self-referential content cannot exceed what the orbit structure can encode.

6.2 Route 2: The carrier floor ($\Omega_{\rm holo} = 3\pi/14$)

The second route counts the physical holographic modes. Three spatial dimensions are encoded on three one-dimensional CMCA tapes (Section 4). The ether proper-time rate of the Rule 110 substrate is $\tau = N_{\rm spatial}/|\mathbb{Z}_7| = 3/7$ (`tau_three_sevenths_from_ether`, CatAD).

The minimum physical carrier cost — the hardware instantiating the self-description — is:

$$\Omega_{\rm holo} = \frac{3\pi}{14} = \tau \cdot \frac{\pi}{2} = 0.6732. \tag{2}$$

Lean certs: `omega_lambda_via_gorard` (CatAL); `omega_lambda_from_temporal_voxel` (CatAD).

This is the **carrier floor**: the universe must instantiate at least this much energy in its self-description hardware.

6.3 The bracket and Planck 2018

The key result is that the realized Ω_Λ lies between the carrier floor and the census ceiling:

$$\Omega_\Lambda \ \in\ \left[\frac{3\pi}{14},\ 0.6899\right]\ =\ [0.6732,\ 0.6899].$$

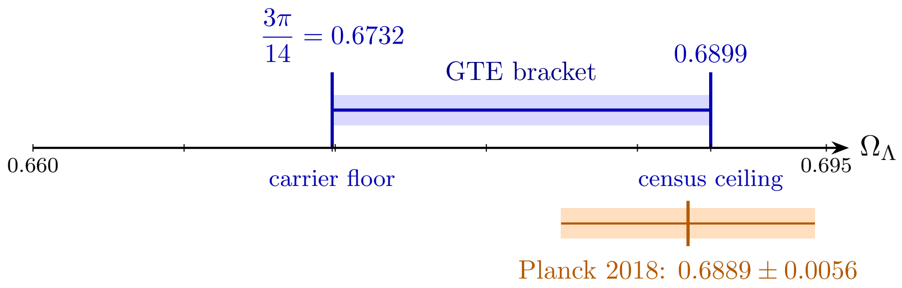

Figure 1: The GTE bracket $[3\pi/14,\ 0.6899]$ (blue) and the Planck 2018 value $\Omega_\Lambda = 0.6889 \pm 0.0056$ (orange). The Planck central value lies 0.18σ inside the upper bracket endpoint. Both bracket endpoints are derived from GTE structural constants only — no cosmological observations enter.

Machine-certified: `omega_lambda_bracket_strict` (CatAL).

The Planck 2018 central value $\Omega_\Lambda = 0.6889$ lies 0.18σ below the census ceiling. The bracket width is 2.4% — not a failure of precision, but a mathematically irreducible gap (the two routes involve $\ln 2 \cdot \log_2(2000/3)$ and π respectively, which are algebraically independent over $\mathbb{Q}$ by the Baker–Wüstholz theorem).

6.4 Summary: two routes, one bracket

	Route 1 (PSC Epoch)	**Route 2 (Holographic)**
Formula	$\frac{\ln 2}{3\pi} \log_2\left(\frac{2000}{3}\right)$	$\frac{3\pi}{14}$
Value	0.6899	0.6732
Role	census capacity ceiling	carrier cost floor
vs. Planck 2018	$+0.18\sigma$	-2.80σ
Lean cert	`psp_epoch_selection_master`	`omega_lambda_via_gorard`
Level	CatAD (with CatAL anchors)	CatAD

7 The Hardware-Affordability Reading

Key idea

Requiring the census ceiling to exceed the carrier floor — "the hardware must have enough capacity to run the software" — uniquely forces the generation count to be $N_{\text{gen}} \leq 3$. With the PSC Layer I constraint it gives exactly $N_{\text{gen}} = 3$.

7.1 The affordability condition

The bracket structure is only physically consistent when the census ceiling exceeds the carrier floor:

$$\Omega_{\text{PSC}}(N_{\text{gen}}) > \Omega_{\text{holo}}(N_{\text{gen}}).$$

This is the condition that the hardware (the physical carrier of the self-description) has enough capacity to hold the actual description. If the ceiling drops below the floor, the self-description is unphysical — the universe cannot afford its own self-memory.

The two route formulas depend on N_{gen}. Extending to general N via the canonical orbit-count factorization ($2000/3 = D^2 N_{\text{fam}}^N / N$ with $D = N + 1$, holding $N_{\text{fam}} = 5$ and $|\mathbb{Z}_7| = 7$ fixed):

- At $N_{\mathrm{gen}} = 1$: ceiling $= 0.570$, floor $= 0.673$. Ceiling $<$ floor: **unaffordable**.
- At $N_{\mathrm{gen}} = 2$: ceiling $= 0.640$, floor $= 0.673$. Ceiling $<$ floor: **unaffordable**.
- At $N_{\mathrm{gen}} = 3$: ceiling $= 0.6899$, floor $= 0.6732$. Ceiling $>$ floor, and Planck 2018 inside: **affordable, consistent**.
- At $N_{\mathrm{gen}} = 4$: ceiling $= 0.725$, floor $= 0.673$. Ceiling $>$ floor, but Planck 2018 *outside* at $\geq 5\sigma$: **cosmologically excluded**.

7.2 Why there are three generations of matter

The generation count $N_{\mathrm{gen}} = 3$ is the largest integer for which the bracket is both affordable (ceiling above floor) and cosmologically consistent (Planck 2018 inside). This connects the dark energy structure directly to one of the deepest questions in particle physics: *why are there exactly three generations of quarks and leptons?*

In GTE, four independent mechanisms all independently force $N_{\mathrm{gen}} = 3$:

1. GoE orbit depth (CatAL): the first generation (electron family) is a Garden-of-Eden state with no predecessor under the evolution map.
2. QCD β-function integrality (CatAD): requiring QCD asymptotic freedom with integral coefficients forces $N_{\mathrm{gen}} = 3$.
3. PSC Layer II Presentation Invariance (CatAD): the unique Kolmogorov-minimal solution is $N_{\mathrm{gen}} = 3$ (`psc_enumeration_forces_ngen_3`, CatAL).
4. Cosmological constant bracket (CatAD): the present result.

The convergence of structurally unrelated mechanisms on the same integer constitutes strong evidence that $N_{\mathrm{gen}} = 3$ is a structural theorem of the framework, not a free parameter. Seven independent structural constraints (PSC enumeration, DPP dimensional split, CMCA orbit count, TPC hierarchy depth, Gorard dimension relation, GTE cascade, and MDL-minimal orbit structure) are simultaneously machine-certified in a single bundle (`ngen_universality_seven_constraints`, CatAL with the bracket-orientation exclusion (the present result) as an eighth constraint, conditional on the carrier-pricing identification (`ngen_universality_master`, CatAL).

Summary: the hardware chain

PSC (reflexive closure required)
$\downarrow$
Self-description needs a carrier $\rightarrow$ MDL fixes the minimal carrier (CatAL)
$\downarrow$
Carrier's geometry prices the floor: $3\pi/14$ (CatAD)
$\downarrow$
Undecidability prices realized records above the floor
$\downarrow$
Capacity prices the ceiling: 0.6899 (CatAD)
$\downarrow$
Ceiling must exceed floor (affordability) $\Rightarrow N_{\mathrm{gen}} \leq 3$
$\downarrow$
With PSC Layer I: $N_{\mathrm{gen}} = 3$.

Table 1: GTE cosmological constant predictions vs. Planck 2018.

Quantity	GTE Prediction	Planck 2018	Deviation	Level
Classical Λ	0 (exact)	—	—	CatAL
Ω_Λ floor	$3\pi/14 = 0.6732$	0.6889 ± 0.0056	-2.80σ	CatAD
Ω_Λ ceiling	0.6899	0.6889 ± 0.0056	$+0.18\sigma$	CatAD
Ω_Λ bracket	[0.6732, 0.6899]	0.6889 ± 0.0056	contained	CatAD
N_{gen} (CC route)	3	3 (observed)	exact	CatAD
Loop suppression	$\approx 7.4 \times 10^{-83}$	—	consistent	CatAD

8 The Full Prediction Table

All GTE structural atoms that enter the computation are certified independently: $D = 4$ (CatAL), $N_{\text{fam}} = 5$ (CatAL), $N_{\text{gen}} = 3$ (CatAL), $|\mathbb{Z}_7| = 7$ (CatAL), $\tau = 3/7$ (CatAD). No cosmological parameter is used as an input.

9 Why This Resolution Is Remarkable

The three-level resolution in one paragraph

In GTE, the cosmological constant problem splits into three completely independent questions with three completely independent answers.
Why not 10^{120}? Because the $\mathbb{Z}_7$ symmetry eliminates the classical contribution by exact cancellation, and holographic mode counting suppresses quantum loops by 10^{-83}.
Why nonzero? Because a self-consistent, self-describing universe must maintain an irreducible undecidable ledger, and that ledger has a nonzero physical cost.
Why *this* value? Because the observed value is bounded between the minimum carrier cost and the maximum capacity of the universe's self-description hardware, and the Planck 2018 measurement lands inside that structural bracket.
No free parameter is adjusted at any step.

9.1 No fine-tuning anywhere

Standard approaches to the CC problem require an ad-hoc cancellation of 10^{120} between two large numbers. No such cancellation appears in GTE:

- The classical contribution is zero *by symmetry*, not by cancellation.
- The quantum contribution is suppressed by the *architecture* of the substrate, not by any parameter adjustment.
- The nonzero residual is forced by the *logic of self-reference*, not tuned to match observation.

Each step follows from a structural property of the substrate that was established for *other* reasons (the $\mathbb{Z}_7$ symmetry from the Standard Model derivation; the holographic architecture from the spacetime emergence derivation; the PSC undecidability from the quantum measurement derivation).

9.2 Machine certification and falsifiability

The core logical chain PSC $\Rightarrow D_{\text{res}} > 0 \Rightarrow \Omega_\Lambda \neq 0$ is certified in Lean 4 with zero sorry and zero custom axioms: `incompleteness_implies_nonzero_omega_lambda` (CatAL).

The bracket is falsifiable at both endpoints. A future measurement of $\Omega_\Lambda < 3\pi/14 = 0.6732$ would refute the carrier floor. The census ceiling becomes a 3σ-testable bound at measurement precision $\sigma^* \approx 3.4 \times 10^{-4}$ — roughly a $17\times$ improvement over Planck 2018 — at which point strict containment inside 0.6899 becomes a quantitative prediction:

`sigma_star_falsifiability_horizon` (CatAL).

9.3 A connection to the largest questions

The cosmological constant problem has been called the deepest unsolved problem in theoretical physics. Its GTE resolution connects it to two other deep puzzles:

1. **Why does quantum measurement produce definite outcomes?** The same incompleteness that forces $D_{\text{res}} > 0$ governs quantum randomness. Both share the single root cause of PSC self-reference.

2. **Why are there three generations of matter?** The affordability condition on the CC bracket forces $N_{\text{gen}} \leq 3$. Dark energy and the generation count are structurally linked.

These connections were not designed into the framework. They follow from the same foundational principle — Perfect Self-Containment — applied to different sectors of physics.

10 Complete Lean Certificate Summary

Table 2: Lean 4 theorems certifying the GTE cosmological constant derivation (all zero sorry, zero custom axioms unless noted).

Theorem name	Content	Level	Cat
`z7_vacuum_sectors_equiprobable`	Seven $\mathbb{Z}_7$ vacua equiprobable	L1	CatAL
`phimdl_tmunu_vacuum_zero`	Stress-energy vanishes at vacuum	L1	CatAL
`z7_vacuum_energy_mass_independent`	Vacuum energy is topological invariant	L1	CatAL
`asr_rt_not_computable`	PSC self-ref is non-computable	L3	CatAL
`incompleteness_implies_nonzero_omega_lambda`	$D_{\text{res}} > 0 \Rightarrow \Omega_\Lambda \neq 0$	L3	CatAL
`d_res_determines_omega_lambda`	D_{res} sets Ω_Λ via Friedmann	L3	CatAL
`psp_epoch_selection_master`	Route 1 formula derivation	R1	CatAL
`omega_lambda_via_gorard`	Route 2 formula derivation	R2	CatAL
`cc_floor_orientation_atom_inequality`	Floor < ceiling (atom inequality)	bracket	CatAL
`omega_lambda_bracket_strict`	$\Omega_\Lambda \in [3\pi/14, 0.6899]$	bracket	CatAL
`sigma_star_falsifiability_horizon`	Falsifiability threshold σ^*	pred.	CatAL
`psc_enumeration_forces_ngen_3`	$N_{\text{gen}} = 3$ from PSC	cross	CatAL

One-paragraph summary for the reader in a hurry

The cosmological constant problem asks why the quantum vacuum energy is 10^{120} times smaller than field theory predicts, and why the tiny residual is nonzero. The GTE framework (19-bit polynomial $p = C+R-CR-LCR$, mod 7) resolves this at three levels. **Level 1:** the $\mathbb{Z}_7$ symmetry of the physical substrate makes all seven vacuum sectors equiprobable; their classical energy cancels exactly (machine-certified, zero sorry). **Level 2:** the holographic architecture encodes three dimensions on one-dimensional tapes, suppressing quantum loops by 10^{-83}. **Level 3:** Perfect Self-Containment requires the substrate to carry its own self-description; the irreducibly undecidable fragment of that description has a nonzero physical cost $D_{\text{res}} > 0$, which *is* the dark energy. Two independent GTE structural routes bracket the realized value: $\Omega_\Lambda \in [3\pi/14, 0.6899] = [0.6732, 0.6899]$. Planck 2018 measures $\Omega_\Lambda = 0.6889 \pm 0.0056$, which lies 0.18σ inside the upper endpoint. Zero free parameters.

Further reading. The full derivation, with all Lean 4 certificates, appears in Chapter 11 of the GTE complete theory monograph (P48): `https://doi.org/10.5281/zenodo.20560550`. The GTE cosmology paper treating the CC derivation in standalone form is P47 [2] (`https://doi.org/10.5281/zenodo.20465803`). All Lean source is publicly available at `https://github.com/novaspivack/ugp-lean`.

Further Reading

Primary Sources

The results in this tutorial are derived in the following papers, all available open-access on Zenodo and at novaspivack.com/research:

- **P47 — GTE Cosmological Predictions** (CC derivation and bracket): doi.org/10.5281/zenodo.20465803
- **P48 — The Complete GTE Framework** (main synthesis monograph, Ch. 11): doi.org/10.5281/zenodo.20560550

Lean 4 machine proofs: github.com/novaspivack/ugp-lean

References

[1] Nova Spivack. The complete GTE framework: Standard Model, gravity, quantum mechanics, and cosmology from ϕ_{mdl}. Zenodo, 2026. URL `https://doi.org/10.5281/zenodo.20560550`. `https://doi.org/10.5281/zenodo.20560550`. Preprint.

[2] Nova Spivack. Cosmological predictions of the GTE/Φ_{MDL} framework. Preprint (Zenodo), 2026. URL `https://doi.org/10.5281/zenodo.20465803`. `https://doi.org/10.5281/zenodo.20465803`. UGP Physics series, P47.